Phase Relation Between the Line Quantities at ω and the DC-Link Voltage Ripple at 2ω in a Single-Phase Inverter

Technical note

August 14, 2026

Abstract

This note derives the exact relation between the phase of the AC-side quantities of a single-phase converter, oscillating at the line angular frequency ω, and the phase of the resulting DC-link voltage ripple at 2ω. The double-line-frequency ripple is shown to be the time integral of the pulsating component of the instantaneous port power, so that its 2ω phasor *leads* the pulsating-power phasor by exactly 90°, and its peak occurs at the line electrical angle $\theta = \varphi/2 - 45°$, where φ is the displacement angle between line voltage and line current. Capacitor ESR, DC-side impedance and control-loop effects are then discussed, together with the consequences for DC-link ripple compensation, feed-forward duty-cycle correction and active power decoupling.

1 Scope and conventions

Consider a single-phase full bridge connected to an AC line through an inductive or LCL filter, and to a DC link consisting of a capacitance C in parallel with a DC source and/or a DC load (the topology is described here verbally: DC source → C → H-bridge → filter → AC line).

The following conventions are used throughout:

- the AC quantities are referred to the *point of connection* of the converter, downstream of the filter, so that the port power is the power actually exchanged with the line;
- power is counted positive when it flows *from the DC link to the AC side* (inverter operation). Rectifier / active-front-end operation is obtained by $\varphi \to \varphi + \pi$, i.e. by a sign reversal of P and Q;
- the line voltage is taken as the phase reference,

$$v(t) = \hat{V}\cos(\omega t) = \hat{V}\cos\theta, \qquad \theta \triangleq \omega t; \tag{1}$$

- the line current lags the voltage by the displacement angle φ,

$$i(t) = \hat{I}\cos(\omega t - \varphi). \tag{2}$$

Switching-frequency components are neglected: only the low-frequency (baseband) behaviour is of interest here.

2 Instantaneous power of a single-phase port

The instantaneous power exchanged with the line is

$$p(t) = v(t)\,i(t) = \hat{V}\hat{I}\cos(\omega t)\cos(\omega t - \varphi) = \frac{\hat{V}\hat{I}}{2}\Big[\cos\varphi + \cos(2\omega t - \varphi)\Big]. \tag{3}$$

Introducing the RMS quantities $V = \hat{V}/\sqrt{2}$, $I = \hat{I}/\sqrt{2}$ and

$$P = VI\cos\varphi, \qquad Q = VI\sin\varphi, \qquad S = VI = \sqrt{P^2 + Q^2}, \tag{4}$$

this becomes

$$\boxed{p(t) = \underbrace{P}_{\text{mean}} + \underbrace{S\cos(2\omega t - \varphi)}_{\tilde{p}(t)}} \tag{5}$$

or, in the equivalent grid-referenced form obtained by expanding the cosine,

$$p(t) = P\big(1 + \cos 2\omega t\big) + Q\sin 2\omega t. \tag{6}$$

Equation (6) is the form to be preferred for implementation, because P and Q are directly available in any grid-connected control structure (DQ or PQ estimator, or the outputs of the outer loops), while φ is not.

Two facts follow immediately and are worth stating explicitly:

1. the amplitude of the pulsating power is the *apparent* power S, not the active power: a converter operating at zero power factor still produces full-amplitude 2ω ripple;
2. the phase of $\tilde{p}$ with respect to the doubled line angle 2θ is $-\varphi$: reactive power *rotates* the pulsating power, it does not attenuate it.

3 DC-link energy balance and voltage ripple

Assume for the moment an ideal DC source delivering the constant power P, an ideal (lossless, zero-ESR) capacitor C, and a lossless bridge. The instantaneous mismatch between the constant power supplied by the source and the pulsating power absorbed by the AC port must be provided by the capacitor:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}Cv_{\mathrm{dc}}^2(t)\right) = P - p(t) = -\tilde{p}(t) = -S\cos(2\omega t - \varphi). \tag{7}$$

Integrating,

$$\frac{1}{2}Cv_{\mathrm{dc}}^2(t) = \frac{1}{2}CV_{\mathrm{dc}}^2 - \frac{S}{2\omega}\sin(2\omega t - \varphi), \tag{8}$$

where V_{dc} is the DC (mean-square) value of the link voltage. Hence the *exact* large-signal solution

$$v_{\mathrm{dc}}(t) = V_{\mathrm{dc}}\sqrt{1 - \frac{S}{\omega CV_{\mathrm{dc}}^2}\sin(2\omega t - \varphi)}\,. \tag{9}$$

The peak-to-peak energy swing of the capacitor is

$$\Delta W_{\mathrm{pp}} = \frac{S}{\omega}, \tag{10}$$

which is the quantity that fixes the minimum storage requirement of any passive or active solution and is independent of the chosen DC voltage.

3.1 Small-ripple linearisation

For $\Delta v_{\mathrm{dc}} \ll V_{\mathrm{dc}}$ (the normal design case, ripple of a few percent), a first-order expansion of (9) gives

$$\boxed{\Delta v_{\mathrm{dc}}(t) \simeq -\frac{S}{2\omega C V_{\mathrm{dc}}}\,\sin(2\omega t-\varphi) = \hat{V}_r \cos\!\Big(2\omega t-\varphi+\frac{\pi}{2}\Big), \qquad \hat{V}_r = \frac{S}{2\omega C V_{\mathrm{dc}}}} \tag{11}$$

or, in the P–Q form,

$$\Delta v_{\mathrm{dc}}(t) \simeq -\frac{1}{2\omega C V_{\mathrm{dc}}}\Big(P\sin 2\omega t - Q\cos 2\omega t\Big). \tag{12}$$

The second-order term neglected in (11) generates a small 4ω component of relative amplitude $\hat{V}_r/(4V_{\mathrm{dc}})$ with respect to the 2ω one; for a 5 % ripple this is about 1.25 % of the 2ω ripple and is normally negligible, but it is the reason why a purely sinusoidal 2ω cancellation never gives exactly zero residual ripple.

4 The phase relation

4.1 Statement of the result

A phase difference between a quantity at ω and a quantity at 2ω is not defined in the usual phasor sense: the two rotate at different speeds. The relation is well posed only if both are expressed as functions of the same angular variable, i.e. of the line electrical angle $\theta = \omega t$ measured from a common time origin (in practice, the output of the PLL). With that convention the results above give

$$\tilde{p}(\theta) = S\cos(2\theta-\varphi), \qquad \Delta v_{\mathrm{dc}}(\theta) = \hat{V}_r\cos\!\Big(2\theta-\varphi+\frac{\pi}{2}\Big). \tag{13}$$

Three equivalent statements of the same result:

In the 2ω domain. Treating 2ω quantities as phasors, the DC-link ripple *leads* the pulsating power by exactly 90°:

$$\Delta\bar{V}_{\mathrm{dc}} = \frac{1}{\mathrm{j}\,2\omega\,C\,V_{\mathrm{dc}}}\,(-\tilde{\bar{P}}) \implies \angle\Delta\bar{V}_{\mathrm{dc}} = \angle\tilde{\bar{P}} + \frac{\pi}{2}. \tag{14}$$

The 90° lead is the composition of the −90° of the integration in (7) with the 180° of the minus sign (the capacitor discharges when the AC side demands more than the average).

Relative to the line voltage. The ripple maxima occur at the line angles

$$\boxed{\theta_{\max} = \frac{\varphi}{2} - 45^\circ \pmod{180^\circ}} \tag{15}$$

and the minima 90° later. Note the factor 1/2: a change $\Delta\varphi$ of the displacement angle shifts the ripple by $\Delta\varphi/2$ in line angle (i.e. by $\Delta\varphi$ in the doubled angle).

Relative to the line current. Since the current peaks at $\theta = \varphi$, the ripple peak occurs at $\theta_{\max} - \varphi = -\varphi/2 - 45^\circ$ with respect to the current peak.

Table 1: Position of the DC-link voltage ripple peak, referred to the positive peak of the line voltage, for different displacement angles.

Operating point	φ	$\theta_{\max}$	Remark
Unity PF, inverting	$0°$	$-45°$	ripple peak 45° *before* the voltage peak
PF $= 0.866$ lagging	$30°$	$-30°$	
Pure inductive Q ($Q > 0$)	$90°$	$0°$	ripple peak *at* the voltage peak, $P = 0$
Unity PF, rectifying (AFE)	$180°$	$45°$	mirror image of the inverting case
Pure capacitive Q ($Q < 0$)	$-90°$	$-90°$	ripple peak at the voltage zero crossing

4.2 Special cases and sanity checks

Table 1 collects the position of the DC-link voltage maximum for some relevant operating points, with the inverter sign convention ($p > 0$ from DC to AC).

The unity-power-factor case deserves a comment because it is the one most often misremembered. At $\varphi = 0$ the instantaneous power is maximum at the voltage peak, so one is tempted to expect the DC-link voltage to be maximum or minimum there. It is neither: at $\theta = 0$ the capacitor is at its *nominal* value and is discharging at maximum rate, because the ripple is the *integral* of the power mismatch, not the mismatch itself. The extremes occur at the zeros of $\tilde{p}$, i.e. 45° away from the voltage peaks and zero crossings.

4.3 Waveforms

Figure 1 shows the normalised waveforms for $\varphi = 0$: line voltage, line current, instantaneous power and DC-link ripple.

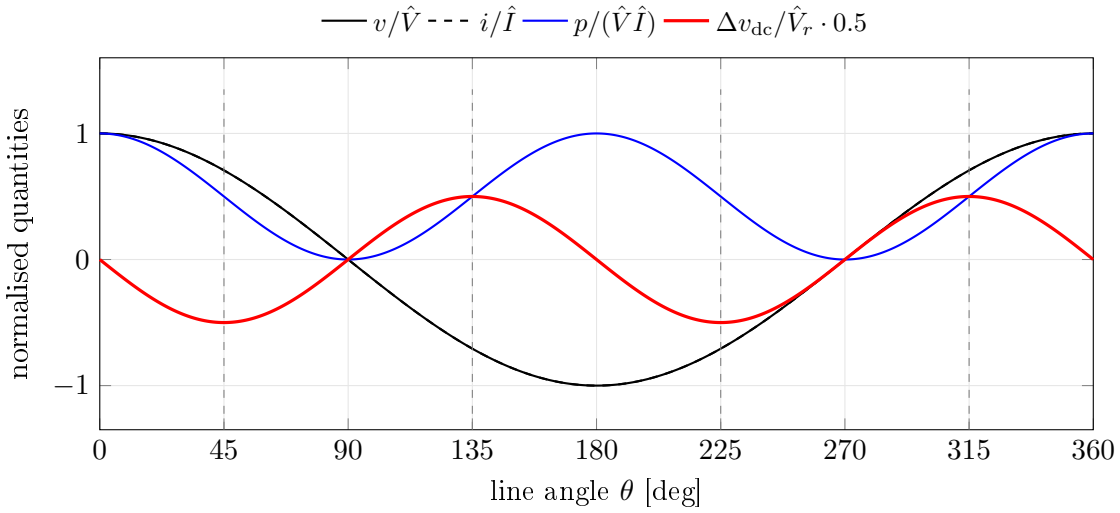


Figure 1: Normalised waveforms for $\varphi = 0$ (unity power factor). The vertical dashed lines mark the extremes of the DC-link ripple, located 45° away from the peaks and the zero crossings of the line voltage. For clarity the ripple is drawn with half amplitude and with the inverter sign convention: at $\theta = 135°$ the capacitor is at its maximum, i.e. it has been charging while $p < P$.

5 Non-idealities that modify the phase

5.1 Capacitor ESR

The current flowing in the DC-link capacitor is, to first order,

$$i_C(t) = \frac{-\tilde{p}(t)}{v_{\mathrm{dc}}(t)} \simeq -\frac{S}{V_{\mathrm{dc}}}\cos(2\omega t - \varphi), \tag{16}$$

so the ESR contribution $R_{\mathrm{esr}} i_C$ is in phase opposition with $\tilde{p}$, while the capacitive contribution leads it by 90°. Their sum gives an additional phase lead

$$\Delta\psi_{\mathrm{esr}} = \arctan\!\big(2\omega C R_{\mathrm{esr}}\big) = \arctan(\tan\delta)_{@2\omega}\,, \tag{17}$$

i.e. exactly the loss angle of the capacitor evaluated at 2ω, and an amplitude increase by the factor $\sqrt{1+(2\omega C R_{\mathrm{esr}})^2}$. For a film capacitor bank at 100 Hz or, worse, at 33.4 Hz, this correction is well below 1° and can be ignored; for large electrolytic banks it can reach a few degrees and, more importantly, is the dominant loss mechanism.

5.2 Finite DC-source and load impedance

Equation (7) assumes that the DC side is a constant-power port at 2ω. In reality the DC source (or the load, e.g. a machine-side inverter) presents a finite impedance and absorbs part of the 2ω current. If $Z_{\mathrm{dc}}(\mathrm{j}2\omega)$ is the equivalent DC-side impedance seen in parallel with C, the ripple becomes

$$\Delta\bar{V}_{\mathrm{dc}} = \frac{-\tilde{\bar{P}}}{V_{\mathrm{dc}}} \cdot \frac{Z_{\mathrm{dc}} \,\|\, \big(1/\mathrm{j}2\omega C\big)}{1}, \tag{18}$$

which reduces to (11) for $|Z_{\mathrm{dc}}| \gg 1/(2\omega C)$. A machine-side inverter operating with a fast torque loop behaves as a constant power load, i.e. as a *negative* resistance at low frequency, which increases the ripple and rotates its phase; this coupling must be checked when the two converters share the same link.

5.3 Voltage-loop reaction

If the outer DC-link voltage regulator has appreciable gain at 2ω, it modulates the active current reference at 2ω, which reacts back on $\tilde{p}$ and hence on the ripple itself. The result is both a change of amplitude and a phase rotation of the ripple, and, on the AC side, a third harmonic of the line current. This is the standard reason for placing a notch at 2ω in the voltage loop, or for using v_{dc}^2 (i.e. the energy) as controlled variable, which also linearises (7) exactly.

6 Consequences for DC-link compensation

6.1 Third harmonic generated by the uncompensated ripple

If the modulator uses a fixed nominal value V_{dc} instead of the measured link voltage, the duty ratio is $d(t) = v^*(t)/V_{\mathrm{dc}}$ and the bridge produces

$$\begin{aligned} v_{\mathrm{out}}(t) &= d(t)\, v_{\mathrm{dc}}(t) = \frac{\hat{V}\cos\theta}{V_{\mathrm{dc}}}\Big[V_{\mathrm{dc}} + \hat{V}_r\cos(2\theta+\psi)\Big] \\ &= \hat{V}\cos\theta + \frac{\hat{V}\delta}{2}\cos(\theta+\psi) + \frac{\hat{V}\delta}{2}\cos(3\theta+\psi), \qquad \delta \triangleq \frac{\hat{V}_r}{V_{\mathrm{dc}}}, \quad \psi = \frac{\pi}{2} - \varphi. \end{aligned} \tag{19}$$

Two effects appear:

- a **third harmonic** of relative amplitude $\delta/2$: a 5 % peak ripple produces 2.5 % of third harmonic voltage, which the filter impedance converts into a current harmonic that is generally not acceptable against IEEE 519 or EN 50160 limits;
- a **fundamental perturbation** of the same amplitude $\delta/2$ and phase ψ, which at unity power factor ($\psi = 90°$) is in quadrature with the reference and therefore appears to the current loop as a spurious reactive component.

The cure is trivial and should always be implemented: divide the voltage reference by the *instantaneous measured* link voltage, $d(t) = v^*(t)/v_{\mathrm{dc}}(t)$, with an adequate measurement bandwidth (at least a decade above 2ω) and taking care of the sampling instant, so that the 2ω content is reconstructed and not aliased.

6.2 Feed-forward reference for an active decoupling stage

An active power decoupling (APD) circuit must absorb $\tilde{p}$ in (5). Its power reference can be built directly from the measured active and reactive power using (6),

$$p^*_{\mathrm{apd}}(t) = \hat{P}\cos 2\theta + \hat{Q}\sin 2\theta, \tag{20}$$

where $\hat{P}$ and $\hat{Q}$ are slow (filtered) estimates and θ comes from the PLL. Equivalently, in energy form, the storage element of the APD must track

$$\tilde{w}^*(t) = \frac{1}{2\omega}\Big(\hat{P}\sin 2\theta - \hat{Q}\cos 2\theta\Big), \tag{21}$$

which is the natural reference when the APD is controlled in the energy variable (i.e. on v^2_{apd}). The feed-forward removes the bulk of the ripple; a resonant (PR) regulator tuned at 2ω then closes the residual error, and the two must be phase-consistent, otherwise the feed-forward degrades rather than improves the result. Equation (15) is the check to be run on the simulation waveforms: with the feed-forward disabled, the residual ripple must peak at $\theta = \varphi/2 - 45°$; if it does not, either the PLL angle, the sign convention or the measurement delay is wrong.

6.3 Passive sizing

Imposing a relative peak ripple $\varepsilon = \hat{V}_r/V_{\mathrm{dc}}$ in (11) gives the classical sizing rule

$$\boxed{C = \frac{S}{2\,\omega\,V_{\mathrm{dc}}^2\,\varepsilon}} \tag{22}$$

which shows the inverse proportionality to ω that makes low-frequency single-phase systems so demanding.

7 Worked example: 350 kV A converter on a 16.7 Hz line

Take $S = 350\,\mathrm{kV\,A}$, $f = 16.7\,\mathrm{Hz}$ ($\omega = 104.9\,\mathrm{rad/s}$, ripple frequency $2f = 33.4\,\mathrm{Hz}$), $V_{\mathrm{dc}} = 700\,\mathrm{V}$, $\varepsilon = 5\,\%$ (i.e. $\hat{V}_r = 35\,\mathrm{V}$, 70 V peak-to-peak).

- Required capacitance, from (22):

$$C = \frac{350\cdot 10^3}{2\cdot 104.9\cdot 700^2\cdot 0.05} \approx 68\,\mathrm{mF}. \tag{23}$$

  For comparison, the same specification at 50 Hz would require 22.8 mF: the 16.7 Hz line costs a factor of three.

- Peak-to-peak energy swing, from (10): $\Delta W_{\mathrm{pp}} = 350 \cdot 10^3/104.9 \approx 3.34\,\mathrm{kJ}$, against a stored energy of $\frac{1}{2}CV_{\mathrm{dc}}^2 \approx 16.7\,\mathrm{kJ}$: only 20 % of the installed energy is actually used, which is the quantitative argument in favour of an active decoupling stage operating with a large voltage swing.
- RMS capacitor current at 2ω: $I_{C,\mathrm{rms}} \approx S/(\sqrt{2}V_{\mathrm{dc}}) = 354\,\mathrm{A}$, which is usually the binding constraint on the bank, not the capacitance itself.
- Phase: at unity power factor the link voltage peaks 45° before the line voltage peak, i.e. $45/(360 \cdot 16.7) \approx 7.5\,\mathrm{ms}$ earlier in time. Note that the same 45° corresponds to 2.5 ms at 50 Hz: when comparing simulations at different line frequencies, always compare angles, never times.
- Uncompensated third harmonic, from (19): $\delta/2 = 2.5\,\%$ of the fundamental voltage.

8 Summary

1. The pulsating power of a single-phase port is $\tilde{p} = S\cos(2\omega t - \varphi) = P\cos 2\omega t + Q\sin 2\omega t$: its amplitude is the apparent power and its phase is $-\varphi$ in the doubled angle.
2. The DC-link ripple is the integral of $-\tilde{p}$ divided by CV_{dc}; therefore its 2ω phasor leads that of $\tilde{p}$ by exactly 90°.
3. Referred to the line electrical angle, the ripple peaks at $\theta = \varphi/2 - 45°$ and its amplitude is $S/(2\omega CV_{\mathrm{dc}})$.
4. A phase difference between ω and 2ω quantities is meaningful only through the common angle θ supplied by the PLL, and is defined modulo 180° in θ.
5. ESR adds a lead equal to the loss angle at 2ω; the DC-side impedance and any voltage-loop gain at 2ω rotate the ripple further and must be included when a 2ω compensator is tuned.

References

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